

Molecular dynamics: 2D Lennard-Jones Simulation

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Implementation We parallelize the algorithm and present several versions that are built on top of each other. We compare the speedup performance with the baseline model. We consider the following 4 improved models:

- **V1 – CPU baseline:** All computation runs sequentially on the CPU. Forces are computed with an $O(n^2)$ nested loop, counting each pair twice.
- **V2 – OpenMP CPU:** Same $O(n^2)$ force computation as V1 but parallelised across CPU cores with OpenMP. Each pair is visited once using Newton’s third law, with per-thread scratch arrays to avoid race conditions.
- **V3 – GPU 2D grid:** Force computation is offloaded to the GPU using a 2D thread grid where each thread handles one (i, j) pair. Integration stays on the CPU, and GPU memory is allocated and freed every step.
- **V4 – GPU 1D pairs:** Force computation uses a 1D kernel with one thread per unique pair and applies Newton’s third law to halve the work. Integration still runs on the CPU with per-step memory transfers.
- **V5 – Fully GPU, AoS:** All steps — forces, integration, and energy computation — run on the GPU. Memory is allocated once and data never leaves the device during the simulation loop.
- **V6 – Fully GPU, SoA:** Identical to V5 but uses a Structure-of-Arrays memory layout, enabling contiguous memory access patterns across threads and reducing the number of force evaluations via r^2 comparisons.
- **V7 – GPU neighbour list:** Introduces a Verlet neighbour list that limits force computation to nearby particles only, breaking the $O(n^2)$ bottleneck. The list is rebuilt only when any particle moves more than half the skin thickness, amortising the rebuild cost over many steps.
- **V8 – 2-GPU naive:** Particles are split evenly across two GPUs, each computing forces for its own half. Position data is exchanged between devices each step via peer-to-peer transfers where available, otherwise staged through the host.

The results show consistent speedups as optimisations are added across all particle counts. V3 is roughly $2\times$ faster than V2, primarily due to halving the number of force evaluations. The largest gain comes from eliminating CPU-GPU transfers in V4, which is $3\text{--}4\times$ faster than V3 at small N and around $2\times$ at $N = 8000$. V5 adds a further $1.03\text{--}1.32\times$ improvement over V4, with the gain growing with N as coalesced memory access becomes more beneficial at larger problem sizes. Overall, V5 achieves a $3.6\text{--}8.1\times$ speedup over V2 depending on particle count. To make results meaningful, we also study the standard deviation. We observed a high standard deviation in V2, which is likely due to variable overhead from repeated memory allocation and deallocation each step.

With respect to the block size, we observe the best configuration across different number of particles is always 256 threads per block.

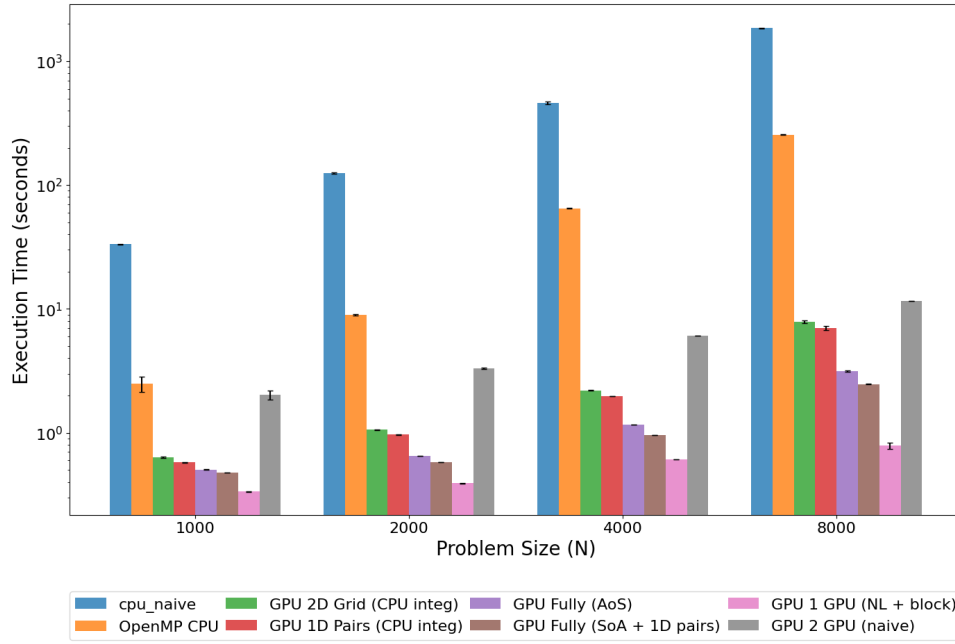


Figure 1: Performance comparison of the aforementioned CPU and CUDA optimization layers.

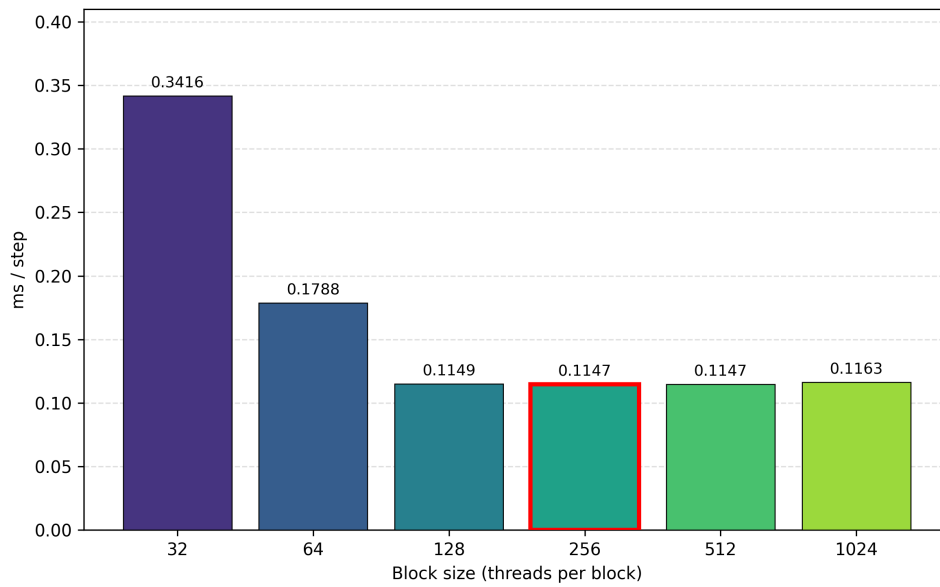


Figure 2: Block size tuning results for the V5 force kernel ($N = 4000$).